

LEGALEASE — DATA FLOW DIAGRAM

LEVEL 0 (Context Diagram):

[User] → (Input: doc type, parties, terms, dates) → [LegalEase System] → (Output: legal document)

LEVEL 1 (Main Components):

User → Streamlit Frontend → HTTP POST → FastAPI Backend → Gemini API → FastAPI Backend → Streamlit Frontend → User

DATA STORES:

- .env file: stores GEMINI_API_KEY
- Session state: stores generated_text and doc_type between Streamlit reruns

DATA FLOW DETAILS:

1. User enters form data in Streamlit → stored in Python variables
2. Streamlit sends POST request to `http://localhost:8000/generate` with JSON payload
3. FastAPI route `/generate` receives `DocumentRequest` (Pydantic model)
4. `GeminiDocumentGenerator.generate_document()` constructs a prompt
5. Prompt sent to Google Gemini API
6. Gemini returns legal text
7. FastAPI returns `{"document": "..."} JSON response`
8. Streamlit displays preview and download buttons